

Churn Reduction Strategy for Company A Telecom

A Data-Driven Business Proposal Using Machine Learning

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Telecom industry & churn trends

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100,000 customer records explored

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Random Forest · AUC 0.673

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Quantified value & roadmap

01 · Market Context

~20–30%

Annual churn rate
in US telecom¹

\$200+

Average cost to
acquire one new customer²

5×

More expensive to acquire
than to retain³

01 · Market Context

Industry Opportunity

- Telecom providers with ML-based churn models report 15–25% reduction in voluntary churn⁴
- Proactive retention campaigns targeted to high-risk customers yield 3–5× ROI vs. broad offers
- Company A's ~49.6% churn rate signals an urgent need for data-driven intervention

02 · Dataset & Exploratory Data Analysis

100,000

Customer records

100

Features

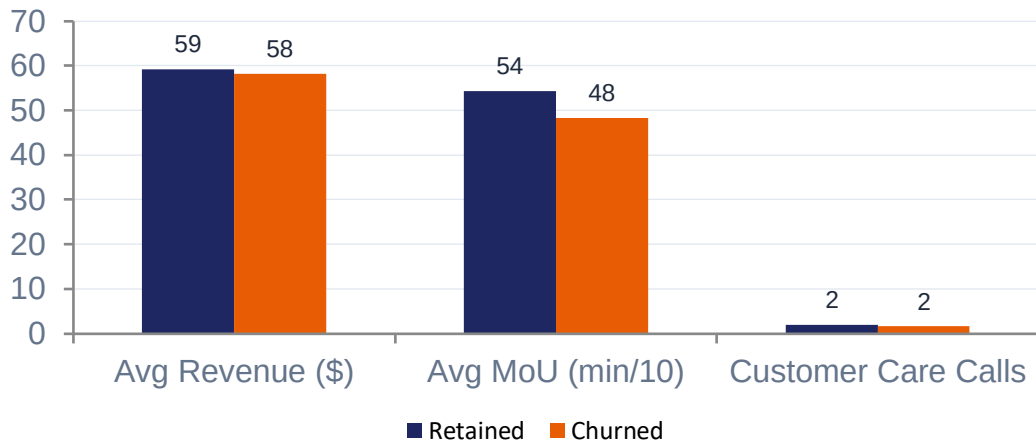
2

CSV tables merged

49.6%

Churn rate

Avg behaviour: Churned vs. Retained



Key EDA Findings

- Churned customers show lower MoU — disengagement precedes exit
- Declining change_mou is a strong pre-churn signal
- Older equipment (high eqpdays) correlates with churn
- Revenue difference is small — price alone doesn't explain churn

02 · Dataset & Exploratory Data Analysis

Company A · 100,000 customers · Telecom churn prediction

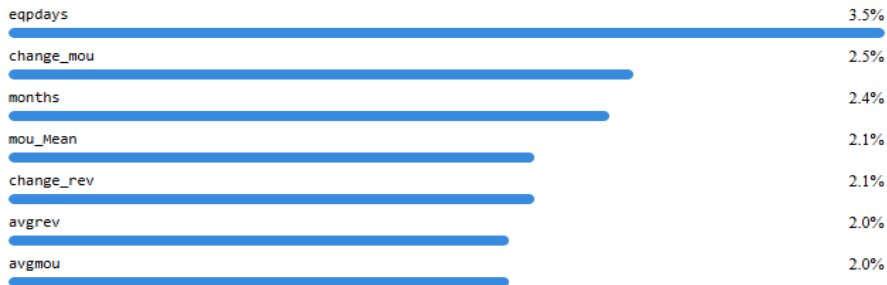
Total customers
100,000

Churn rate
49.6%

Model (random forest)
AUC 0.673

F1-score
0.62

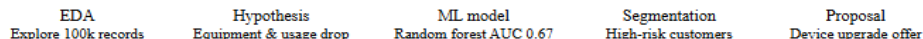
Top churn predictors



Churned vs retained: avg behaviour



Proposed business workflow



Key insight 1

Equipment age (eqpdays) is the #1 predictor — customers with old devices churn more

Key insight 2

Declining MoU (change_mou) strongly signals disengagement before churn event

Proposal

Proactive device upgrade + retention offer triggered at high-risk score threshold

03 · Machine Learning Model

Model

Random Forest Classifier

0.673

ROC-AUC

0.62

F1-Score

0.62

Precision

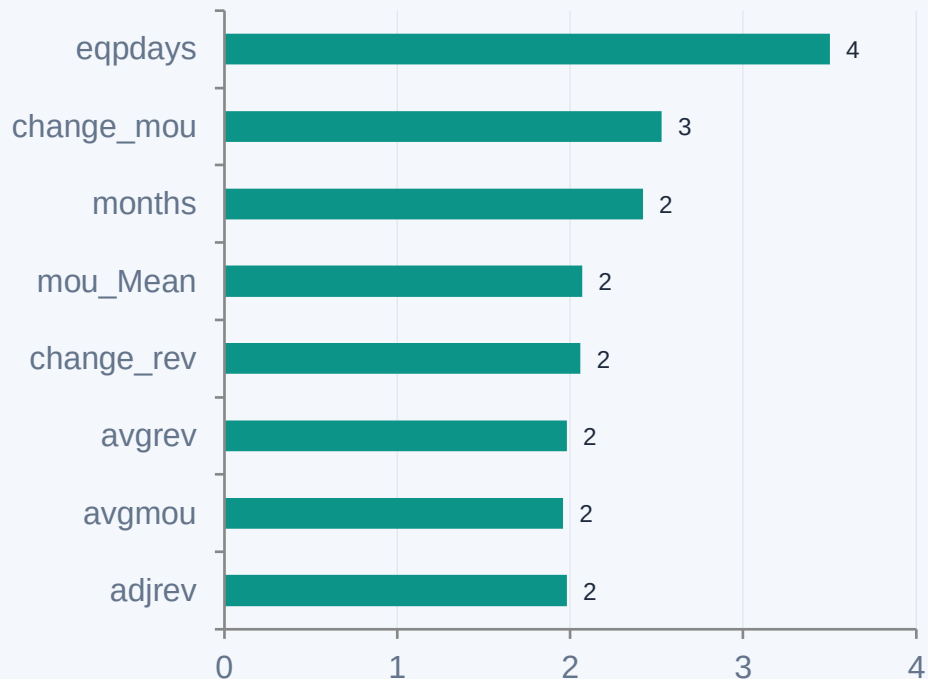
0.61

Recall

n_estimators=100 · test_size=20%
Label-encoded categoricals · median imputation

03 · Machine Learning Model

Top Churn Predictors (Feature Importance)



04 · Key Findings

01 Equipment Age is the #1 Driver

eqpdays (age of current handset) has the highest feature importance at 3.5%. Customers using older devices are significantly more likely to churn — hardware obsolescence creates friction.

02

Revenue Alone Doesn't Predict Churn

rev_Mean difference between churned (\$58.21) and retained (\$59.22) is minimal — high-value customers churn too. Generic discount campaigns will not solve this problem.

04 · Key Findings

04

Tenure Matters, But Isn't Enough

months in service ranks 3rd. Both short-tenure (onboarding risk) and long-tenure (entrenched habits) customers churn. Retention strategy must be lifecycle-aware.

03

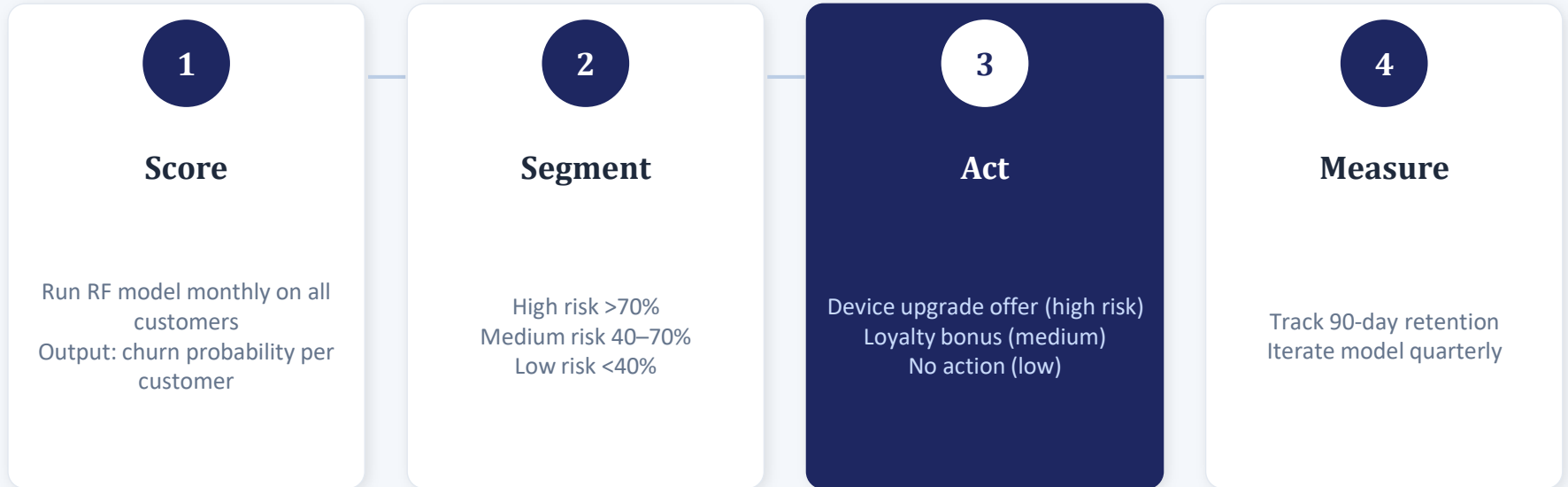
Declining Usage Precedes Churn

change_mou (MoU trend vs prior 3-month avg) is the 2nd strongest predictor. A sustained drop in call minutes is a detectable early warning sign — up to 30–60 days before churn.

05 · Business Proposal

Proactive AI-Powered Retention System

Deploy a monthly scoring pipeline that identifies high-risk customers before they churn and triggers personalised retention actions.



06 · Estimated Impact & Next Steps

~7,440

Customers saved
(10% churn reduction)

\$5.2M

Annual revenue
retained (est.)

3–5×

Expected ROI
on retention spend

Assumptions: avg monthly revenue \$58.72/customer · 100,000 customers · 10% churn reduction scenario

06 · Estimated Impact & Next Steps

Recommended Next Steps

Month 1–2

Deploy pilot

Run model on 20% of customers; set up A/B test with and without retention offers

Month 3–4

Refine model

Add new features (contract type, network coverage); test XGBoost & neural net variants

Month 5–6

Full rollout

Scale to 100% of base; integrate into CRM for real-time scoring; report ROI to stakeholders

References

1. CTIA – The Wireless Association (2024). CTIA Annual Survey. <https://www.ctia.org/>
2. Bain & Company (2020). Customer Acquisition Model.
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4. McKinsey & Company (2017). Reducing churn in telecom through advanced analytics.
5. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.
6. GCI World 2026 April – Company A Dataset Overview (provided dataset documentation).